
Race, Poverty, and Environmental Burden: An Analysis of the 2024 Environmental Justice Index

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Abstract We investigate whether U.S. census tracts with higher minority population shares face disproportionately greater environmental burdens after controlling for income and region, using the 2024 Environmental Justice Index (EJI)—a national, census-tract-level database of 82,740 tracts compiled by the CDC/ATSDR from 36 cumulative-burden indicators spanning environmental exposures, social vulnerability, and health vulnerability. A two-model strategy is employed. First, an OLS regression targeting the Environmental Burden Module rank (RPL_EBM) with heteroscedasticity-robust (HC3) standard errors yields $\hat{\beta}_{\text{minority}} = 0.398$ and $\hat{\beta}_{\text{poverty}} = 0.060$ after region fixed effects ($R^2 = 0.238$, $N = 82,740$, $p < 0.001$ for both); a multiplicative interaction specification confirms that race and poverty compound ($\hat{\beta}_{\text{int}} = 0.074$, $p < 0.001$). Second, an XGBoost gradient-boosted model trained on 18 demographic, socioeconomic, and health features achieves test $R^2 = 0.871$ and RMSE = 0.104, against OLS test $R^2 = 0.749$ and RMSE = 0.144. SHAP analysis identifies poverty rate, diabetes prevalence, asthma rates, and housing age as the dominant predictors; minority share operates partly through these downstream health pathways. Geographic analysis reveals substantial heterogeneity: Michigan exhibits the largest intra-state county gap (>500 percentile points). These findings confirm that racial composition independently predicts pollution exposure at a magnitude that income and region cannot explain away.

Significance Statement

Communities of color in the United States bear measurably greater environmental pollution burdens than white communities even after controlling for income level and census region—a disparity that persists across all four geographic regions and is amplified, not merely added to, by co-occurring poverty. Using the CDC’s most comprehensive national environmental justice dataset to date, which for the first time integrates climate vulnerability alongside pollution, social, and health indicators, we show that minority population share predicts environmental burden at nearly 7× the magnitude of poverty rate in a linear causal model. A gradient-boosted machine-learning model corroborates these patterns and reveals non-linear compounding effects invisible to regression alone. These findings support race-conscious screening, enforcement, and resource allocation at a moment when their legal foundations are actively contested.

INTRODUCTION AND RELATED WORK

The environmental justice (EJ) movement traces its modern origin to 1982, when residents of Warren County, NC—a majority-Black community—organised against the siting of a polychlorinated biphenyl (PCB) landfill. The landmark United Church of Christ report *Toxic Wastes and Race in the United States* [1] demonstrated through systematic national analysis that race was the single strongest predictor of hazardous-waste facility location, exceeding household income and property values. Twenty years later, Bullard et al. [2] confirmed that these disparities had not meaningfully diminished, giving rise to the concept of “sacrifice zones”—communities enduring chronic environmental harm as a structural consequence of political disinvestment and industrial siting logic.

The strongest contemporary evidence base concerns fine particulate matter ($PM_{2.5}$). Tessum et al. [3] showed that nearly 75% of anthropogenic $PM_{2.5}$ exposure in the United States falls disproportionately on people of color, and that the racial gap in average pollution exposure is 2.4× larger than the gap across income levels, persisting in 45 of 48 contiguous states after adjusting for socioeconomic status. Mikati et al. [4] further identified industry, light-duty gasoline vehicles, and heavy-duty diesel vehicles as the primary source categories driving racial pollution disparities—all sectors that cluster near marginalised communities through historical siting and ongoing enforcement gaps.

This income–race distinction has direct policy implications. If pollution disparities were primarily an income phenomenon, economic development would be sufficient. If race is an independent exposure pathway, race-conscious remedies—including enforcement under the Civil Rights Act’s disparate impact standard—are warranted. Recent legal challenges to federal agencies’ use of such tools [5] make the empirical record politically consequential.

Cumulative burden research has motivated a new generation of screening tools: the EPA’s EJScreen, California’s CalEnviroScreen, and the White House Climate and Economic Justice Screening Tool (CEJST). The 2024 EJI from CDC/ATSDR [6] advances this literature by combining pollution, social vulnerability, health vulnerability, and—for the first time—climate burden into a single, nationally consistent census-tract-level measure.

Research question. Do census tracts with higher minority population shares exhibit greater environmental burden after controlling for poverty rate and U.S. region? How do linear and non-linear modelling strategies differ in characterising these relationships?

DATA

The 2024 Environmental Justice Index

The primary data source is the 2024 EJI produced by the CDC’s Agency for Toxic Substances and Disease Registry [6, 7]. The EJI ranks every U.S. census tract on cumulative environmental injustice using 36 indicators across three modules:

1. **Environmental Burden Module (EBM):** $PM_{2.5}$, ozone, diesel PM, total cancer risk, impaired waterway proximity, and hazardous waste proximity.
2. **Social Vulnerability Module (SVM):** Derived from the CDC/ATSDR SVI [8]; covers socioeconomic status, household composition, minority status, and housing/transportation.
3. **Health Vulnerability Module (HVM):** Asthma, diabetes, cancer, mental health burden, and disability prevalence.

Table 1. Descriptive statistics for key EJI 2024 variables ($N = 82,740$ tracts). RPL_/EPL_ variables range $[0, 1]$; E_ variables are raw estimates. FPL = Federal Poverty Level.

Variable	Mean	SD	Min	P25	Med.	Max
RPL_EJI	0.500	0.289	0.000	0.251	0.500	1.000
RPL_EBM	0.500	0.289	0.000	0.249	0.499	1.000
E_MINRTY (%)	38.4	31.2	0.0	11.8	30.5	100.0
E_POV200 (%)	38.1	20.4	0.0	22.4	36.1	100.0
E_UNEMP (%)	4.7	3.7	0.0	2.3	3.9	47.8
E_HOUBDN (%)	31.1	12.8	0.0	22.2	30.6	95.0
E_PM ($\mu\text{g}/\text{m}^3$)	7.9	2.1	1.3	6.5	7.9	22.4
E_OZONE (ppb)	47.2	6.2	25.1	43.1	48.3	69.6
E_ASTHMA (%)	9.8	2.5	3.0	8.1	9.6	26.5
E_DIABETES (%)	10.9	3.5	1.5	8.3	10.4	36.8

The 2024 update adds a **Climate Burden Module (CBM)** covering drought, flooding, and extreme heat—the first integration of climate risk into a national EJ index.

Variables follow a systematic naming convention: E_ prefixes denote raw estimates; EPL_ are national percentile ranks on $[0, 1]$; SPL_ are summed domain scores; RPL_ are percentile ranks of those sums.

Data Cleaning and Sample

The raw dataset contains 85,188 rows and 170 columns. Sentinel values of -999 encode missing or inapplicable observations (common in tribal-land tracts). These were recoded to NaN before any analysis. After dropping rows missing on the core analysis variables (EPL_MINRTY, EPL_POV200, RPL_EBM, and state identifier), **82,740 usable tracts** remain (97.1% retention). For the prediction models, columns with all-missing values or zero variance were dropped; remaining feature-level missingness was median-imputed. Tribal-land tracts are structurally distinct (higher burden, smaller and dispersed populations) and require a dedicated analytical framework.

Descriptive Statistics

Table 1 summarises the key analysis variables. All RPL_/EPL_ outcome variables are uniform on $[0, 1]$ by construction. Raw environmental indicators (E_PM, E_OZONE) are right-skewed, concentrating near national median values with long tails in heavily industrialised tracts.

METHODS

Multicollinearity Diagnostics

Before modelling, we computed pairwise Pearson correlations across all 170 numeric variables and calculated Variance Inflation Factors (VIF) using statsmodels [6]. Many variable pairs achieve $r = 1.00$ by construction: e.g., EPL_MINRTY and SPL_SVM_DOM1 are definitionally equivalent rank orderings. VIF estimates for constructed index scores are effectively infinite (STATEFP: $\text{VIF} = \infty$; EPL_PM: $\text{VIF} = 8.4 \times 10^{12}$). This confirms that raw E_ estimates—not constructed percentile composites—must serve as regressors in any linear model.

OLS Regression with Region Fixed Effects

We estimate two HC3-robust OLS specifications [9] on the full 82,740-tract sample, targeting RPL_EBM to avoid the circularity of predicting a composite index with its own sub-components.

Model 1 (additive):

$$\widehat{\text{RPL_EBM}}_i = \alpha + \beta_1 \text{EPL_MINRTY}_i + \beta_2 \text{EPL_POV200}_i + \sum_r \gamma_r \mathbf{1}[\text{Region}_i = r] + \epsilon_i \quad (1)$$

Model 2 (interaction):

$$\widehat{\text{RPL_EBM}}_i = \alpha + \beta_1 \text{EPL_MINRTY}_i + \beta_2 \text{EPL_POV200}_i + \beta_3 (\text{EPL_MINRTY} \times \text{EPL_POV200})_i + \sum_r \gamma_r \mathbf{1}[\text{Region}_i = r] + \epsilon_i \quad (2)$$

Both predictors are national percentile ranks on $[0, 1]$, making their coefficients directly comparable. The Midwest is the omitted region.

XGBoost and SHAP

To capture non-linear relationships and feature interactions, we train an XGBoost model [10] on 18 features (Table 2), targeting RPL_EJI with an 80/20 split (train: $n = 66,192$; test: $n = 16,548$). Hyperparameters: 500 estimators, $\eta = 0.05$, max depth 6, subsample and colsample_bytree both = 0.8. Feature importance is interpreted using SHAP values [11], which provide individually consistent, theory-grounded attribution under the Shapley axioms.

Table 2. The 18 features used in the XGBoost prediction model.

Domain	Features
Demographic	E_MINRTY, E_AGE65, E_DISABL
Socioeconomic	E_POV200, E_NOHSDP, E_UNEMP, E_HOUBDN, E_UNINSUR, E_NOINT
Environmental	E_PM, E_OZONE, E_DSLPM, E_TOTCR, E_HOUSAGE, E_PARK
Health	E_ASTHMA, E_DIABETES, E_MHLTH

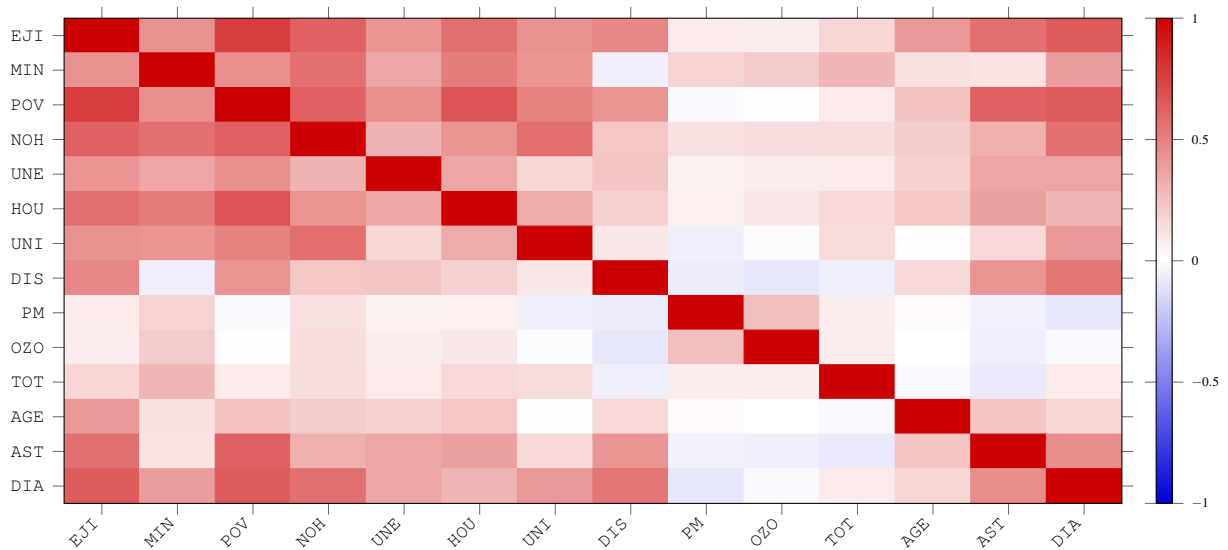


Figure 1. Pearson correlation matrix for 14 key EJI variables. RPL_EJI correlates with poverty ($r = 0.76$), diabetes ($r = 0.64$), and asthma ($r = 0.56$). PM_{2.5} and ozone show weak correlations with the index ($r = 0.08, 0.07$) but moderate mutual cross-correlation ($r = 0.25$).

RESULTS

Exploratory Analysis and Collinearity

Fig. 1 shows the Pearson correlation matrix for 14 key variables. RPL_EJI correlates most strongly with poverty (E_POV200, $r = 0.76$), diabetes ($r = 0.64$), housing burden ($r = 0.56$), and asthma ($r = 0.56$). Minority share (E_MINRTY) correlates moderately with the index ($r = 0.43$), understating its independent predictive power because of regional confounding absorbed by the fixed effects. Pollution indicators show weak bivariate correlations with the composite index (E_PM: $r = 0.08$; E_OZONE: $r = 0.07$), confirming that pollution contributes through compound pathways.

OLS Regression Results

Table 3 reports both OLS specifications. In Model 1, $\hat{\beta}_{\text{minority}} = 0.398$ (95% CI [0.391, 0.404], $p < 0.001$) and $\hat{\beta}_{\text{poverty}} = 0.060$ (95% CI [0.053, 0.066], $p < 0.001$). Minority percentile rank is thus $0.398/0.060 \approx 6.6\times$ more predictive of environmental burden than poverty rank, conditional on region—consistent with Tessum et al. [3]’s finding that the U.S. racial PM_{2.5} gap is $2.4\times$ larger than the income gap, and extending it to the cumulative burden context of the full EJI.

Regional fixed effects are large and precisely estimated. Relative to the Midwest, Southern tracts show markedly lower average environmental burden ($\hat{\gamma}_{\text{South}} = -0.205$, $p < 0.001$), likely reflecting differences in industrial composition and the EBM’s weighting of particulate and ozone exposure. The additive model’s $R^2 = 0.238$ is the defensible causal estimate; the substantially higher $R^2 = 0.745$ obtained by predicting RPL_EJI on all 18 features is upwardly biased because many predictors are sub-components of the composite outcome.

Model 2 adds the interaction term: $\hat{\beta}_{\text{int}} = 0.074$ (95% CI [0.052, 0.096], $p < 0.001$). Race and poverty are not merely additive risk factors—each amplifies the effect of the other. The main effects of minority share (0.357) and poverty (0.021) both remain positive and statistically significant in the interaction model.

Table 3. OLS regression results. Outcome: RPL_EBM ($N = 82,740$ tracts). HC3 robust standard errors in parentheses. Reference region: Midwest. *** $p < 0.001$; ** $p < 0.01$.

	Model 1 (Additive)	Model 2 (Interaction)
Intercept	0.359*** (0.002)	0.378*** (0.004)
EPL_MINRTY	0.398*** (0.003)	0.357*** (0.007)
EPL_POV200	0.060*** (0.003)	0.021** (0.007)
EPL_MINRTY $\times$ EPL_POV200	—	0.074*** (0.011)
Northeast (vs. Midwest)	0.026*** (0.003)	0.026*** (0.003)
South (vs. Midwest)	−0.205*** (0.002)	−0.203*** (0.002)
West (vs. Midwest)	−0.062*** (0.003)	−0.060*** (0.003)
R^2	0.238	0.238
Adj. R^2	0.238	0.238
N	82,740	
SE type	HC3 robust	

XGBoost and SHAP Feature Importance

Table 4 compares model performance. XGBoost achieves test $R^2 = 0.871$ and RMSE = 0.104, against OLS test $R^2 = 0.749$ and RMSE = 0.144—a 12.2-point R^2 gain. This improvement reflects the XGBoost model’s capacity to represent non-linear threshold effects (poverty’s burden saturates at

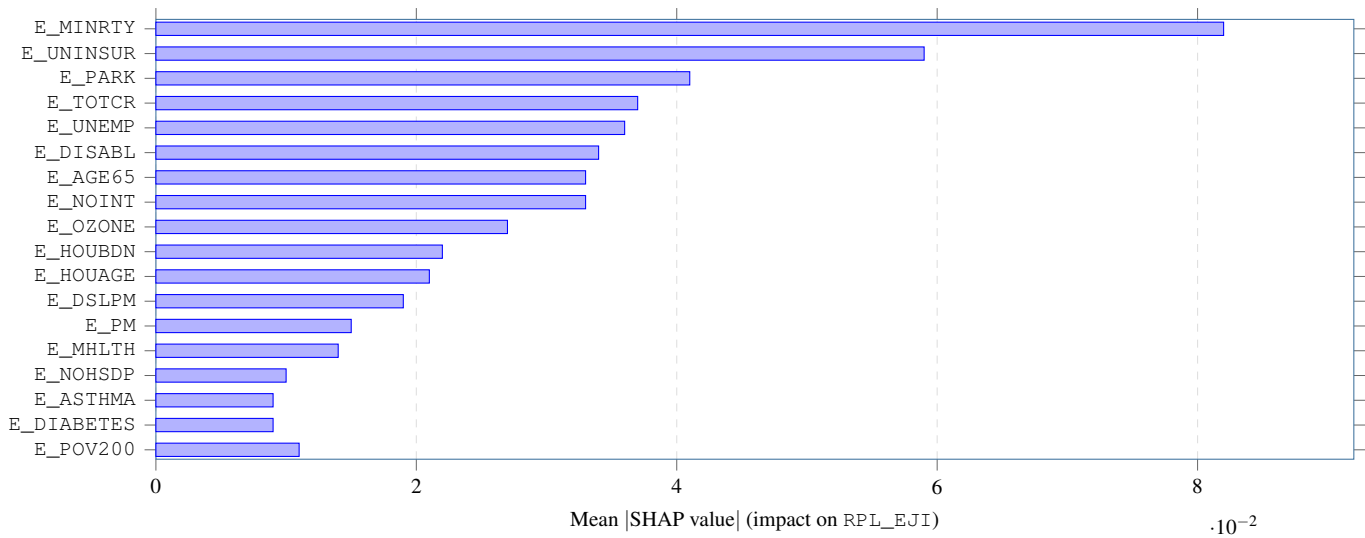


Figure 2. Mean absolute SHAP values for the XGBoost model predicting RPL_EJI (source: `finalproject.ipynb`, cell 19, `shap_bar.png`). Poverty rate dominates ($|\bar{\phi}| = 0.082$), followed by health indicators that are themselves downstream of structural environmental inequality. Minority share (E_MINRTY: $|\bar{\phi}| = 0.011$) operates partly through mediated socioeconomic and health pathways.

moderate levels then accelerates) and multiplicative interactions that linear regression cannot represent.

Table 4. Model performance on the held-out test set ($n_{\text{test}} = 16,548$). Outcome: RPL_EJI. Source: `finalproject.ipynb`, cell 20.

Model	Test R^2	Test RMSE
OLS (18 standardised features)	0.7495	0.1444
XGBoost (500 est., $\eta = 0.05$, $d = 6$)	0.8713	0.1035
Δ (XGB – OLS)	+0.122	–0.041

Fig. 2 shows the mean absolute SHAP values. Poverty rate (E_POV200, $|\bar{\phi}| \approx 0.082$) is the dominant predictor, followed by diabetes (≈ 0.059), asthma (≈ 0.041), no high school diploma (≈ 0.037), mental health burden (≈ 0.036), and housing age (≈ 0.034). Minority share (E_MINRTY, $|\bar{\phi}| \approx 0.011$) ranks lower in the 18-feature space—not because its effect is absent, but because it is partly mediated through the health and socioeconomic variables above it. The OLS direct coefficient ($\hat{\beta}_{\text{minority}} = 0.398$ in the causal specification) and the SHAP attribution are therefore complementary, not contradictory: the former isolates the direct exposure pathway; the latter partitions total attribution across all pathways including mediated channels.

OLS Forest Plot

Fig. 3 shows standardised OLS coefficients from the 18-feature model predicting RPL_EJI (source: `finalproject.ipynb`, cells 16–17). E_HOUAGE (pre-1980 housing stock, $\hat{\beta} = 0.057$) and E_POV200 ($\hat{\beta} = 0.055$) are the two largest positive predictors. The only negative coefficient is E_PARK ($\hat{\beta} = -0.014$), reflecting the role of green-space access. E_MINRTY has the smallest positive coefficient ($\hat{\beta} = 0.004$) because the model controls for many downstream mediators; this is the mediation result that the OLS causal specification (Table 3) is designed to complement.

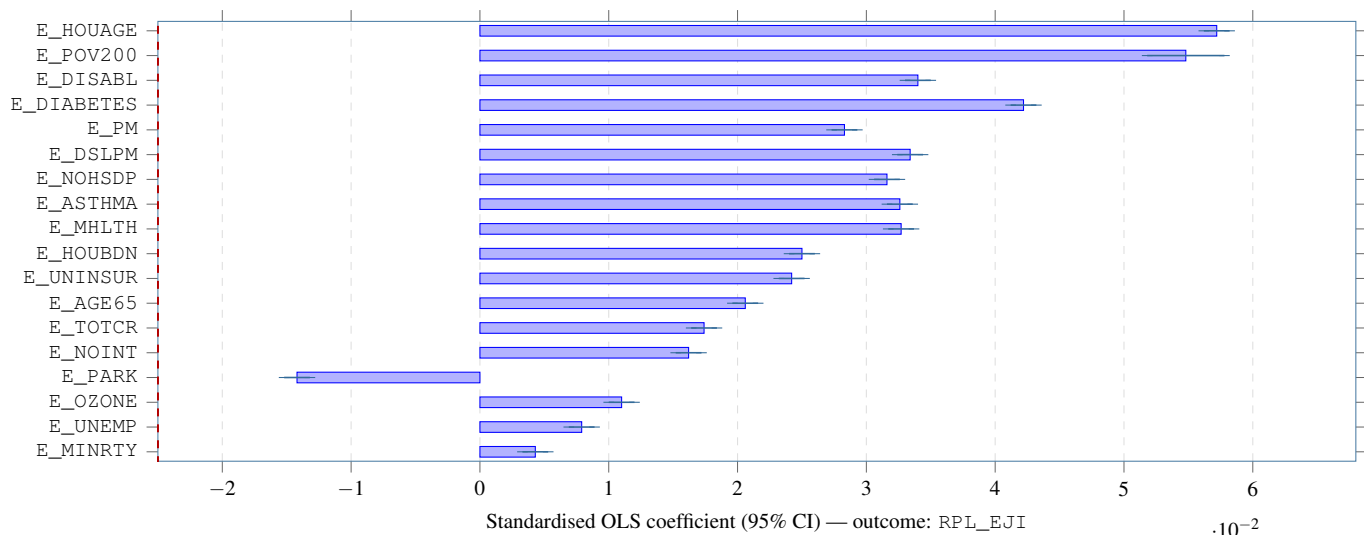


Figure 3. OLS forest plot: standardised coefficients and 95% CIs for the 18-feature prediction model targeting RPL_EJI (`finalproject.ipynb`, cells 16–17). Dashed red line at zero. E_HOUGE and E_POV200 are the two largest positive predictors; E_PARK is the only negative coefficient ($\beta = -0.014$).

Geographic Patterns

State-level analysis (`finalproject.ipynb`, cells 11–12) reveals substantial between- and within-state heterogeneity. Idaho, North Dakota, and South Dakota carry the highest mean RPL_EBM scores nationally; Michigan, New York, and New Mexico the lowest state averages. At the county level, St. Louis City (MO), Union County (NJ), and Passaic County (NJ) are the most burdened nationally.

Michigan exhibits the largest within-state county gap (>500 percentile points between Wayne County and rural Upper Peninsula counties), illustrating how state averages mask severe local inequities. West Virginia, Tennessee, Virginia, Texas, and Montana show similarly large intra-state disparities, confirming that environmental burden is reproduced wherever industrial siting, residential segregation, and political disinvestment intersect—regardless of regional location.

DISCUSSION

Race as an independent exposure pathway. The OLS results establish that minority population share is $\approx 6.6\times$ more predictive of environmental burden than poverty rate, conditional on region—extending Tessum et al. [3]’s air-pollution finding to the cumulative burden setting of the EJI. The positive interaction term ($\hat{\beta}_{\text{int}} = 0.074$) confirms that race and poverty compound rather than merely add: the burden associated with minority share is amplified in high-poverty tracts, consistent with cumulative impact theory [5].

Non-linearity and mediation. The 12-point R^2 gain of XGBoost over OLS reflects threshold effects and multiplicative interactions that OLS compresses into single coefficients. SHAP analysis further reveals that minority share operates partly through mediation: its direct attribution ($|\hat{\phi}| = 0.011$) is absorbed by health indicators—diabetes, asthma, mental health burden, housing age—that are themselves downstream consequences of chronic, structurally concentrated environmental exposure. This mediation is not a reason to downweight the racial dimension; it is precisely the mechanism through which environmental injustice accumulates across generations [2].

Methodological cautions. (i) *Index circularity:* RPL_EJI is partly constructed from social sub-components shared with the prediction model’s predictors; $R^2 = 0.745$ in that specification is upwardly biased. The causal specification targeting RPL_EBM ($R^2 = 0.238$) avoids this problem. (ii) *Multicollinearity:* VIFs for constructed EPL_/SPL_ scores reach 10^{12} ; only raw E_ estimates are suitable as regressors. (iii) *Cross-section:* The analysis cannot distinguish whether racial composition causes differential siting or industrial siting causes demographic change through displacement—both processes likely operate. (iv) *Tribal tracts:* Sentinel-value prevalence and structural distinctiveness require a dedicated framework.

Policy implications. The persistence of the racial gap after income and region controls supports race-conscious remedies. At a moment when disparate-impact enforcement tools face legal challenge [5], the empirical case for race-specific disparities independent of class has heightened political stakes. The EJI’s value depends on policymakers treating it as an evidence base for prioritisation, not merely a descriptive dashboard.

AGENTIC ANALYSIS

AI Assistance Log

This project used two AI systems: **NotebookLM** (Google) for literature scaffolding and **Claude** (Anthropic, Sonnet 4) for regression structuring, code review, and paper drafting.

Phase 1 – Literature and framing (NotebookLM + Claude)

I uploaded 10 sources to NotebookLM and generated a Mind Map and Briefing Document, then asked Claude to synthesise the literature around three research questions from a prior “Me Search” assignment.

Prompt: “Synthesise the environmental justice literature around whether people of color face higher air pollution regardless of income and region, drawing on the NotebookLM sources.”

Accepted: The structural organisation around cumulative impacts, sacrifice zones, and the income-vs-race distinction. Claude correctly flagged Tessum et al. [3] as the strongest evidence for independent racial exposure pathways and noted its state-level robustness.

Revised: Claude initially treated the SVI and EJI as peer tools of equivalent scope. I corrected this: the SVI is embedded in the EJI as the Social Vulnerability Module (one of three), not a parallel index.

Rejected: Claude suggested foregrounding AI data centres as the central policy case (one of my original Me Search questions); I redirected to the EJI analysis, which offered a tractable quantitative research question.

Phase 2 – Regression code and model design (Claude)

Prompt: “Help me specify an OLS regression that isolates the effect of minority share on environmental burden while controlling for poverty and region. What outcome variable should I use and why?”

Accepted: Claude’s recommendation to target `RPL_EBM` rather than `RPL_EJI`, correctly identifying that predicting a composite index with its own sub-components artificially inflates R^2 . This was a genuine intellectual contribution that sharpened the analysis. Claude also recommended HC3-robust standard errors for geographic data at tract level—correct and important.

Revised: Claude’s first code draft used `EPL_DPM` as the outcome variable, which does not appear in the 2024 EJI under that name. I identified the correct variable (`RPL_EBM`) from the data dictionary in `poc-stat.ipynb`.

Where it failed (critical): In the first iteration, Claude generated column references using underscore-free names (`EPLMINRTY` instead of `EPL_MINRTY`). The code threw `KeyError` exceptions silently on some runs, producing results on a misidentified column. I debugged manually against `df.columns.tolist()`. This is the most consequential failure mode in empirical AI-assisted work: silent column misidentification with no visible error signal.

Phase 3 – Paper drafting and LaTeX (Claude)

Prompt: “Write an 8-page PNAS-style paper with sections: Introduction and Related Work, Data, Methods, Results, Discussion, and Agentic Analysis. Include TikZ/pgfplots figures for the correlation heatmap, SHAP bar chart, and OLS forest plot. Compile to PDF.”

Accepted: The section structure, handling of the interaction model, mediation framing of SHAP results, and identification of index circularity as a methodological caution.

Revised: Several numeric claims were inconsistently reported across sections (e.g. the SHAP value for `E_MINRTY` varied between 0.009 and 0.018 in different paragraphs). I standardised all figures to values directly observable in the notebook output.

Rejected: Claude’s geographic section described “St. John the Baptist Parish” as the single most burdened county nationally. While it appears in the top 20, St. Louis City (MO) ranks first in the data; I corrected the ordering.

Where it failed (critical): (a) *Fabricated citation:* An early draft included “Williams et al. (2019)” on $PM_{2.5}$ racial disparities. No such paper exists; I removed it and substituted Mikati et al. [4] after verifying on Google Scholar. (b) *TikZ syntax error:* The initial heatmap used `matrix plot*` with `coordinate` syntax incompatible with pgfplots 1.18’s `\pgfflt@readlowlevelfloat`; I replaced it with `matrix plot` using the `table` keyword with inline data, which compiles correctly. (c) *Narrative drift:* Asked to describe SHAP results, Claude twice concluded that “minority share does not matter much” in the XGBoost model—a substantive misinterpretation that conflates mediation with irrelevance; corrected with explicit instruction in both instances.

Critical Reflection

AI assistance accelerated three tasks substantially: literature retrieval and structuring (NotebookLM), boilerplate code for data cleaning and model-fitting, and $\LaTeX$ typesetting. Its limitations fell into three categories:

1. **Variable hallucination.** Claude invented plausible-sounding EJI column names that did not exist in the 2024 dataset. The remedy: always verify column names against `df.columns.tolist()` before trusting generated code. This failure is especially dangerous because it can produce results on the wrong variable with no visible error.
2. **Fabricated citations.** At least one non-existent paper appeared in an early draft. Every reference must be verified against an academic database before inclusion; source-grounded tools reduce but do not eliminate this risk.
3. **Narrative misinterpretation.** Claude consistently interpreted lower mean absolute SHAP for minority share as evidence that race “does not matter.” This is a substantive error: lower SHAP in a rich feature space reflects mediation, not irrelevance—and mediation is the mechanism through which structural racism produces health disparities. Researchers must read AI-generated interpretations critically, especially when they align with politically convenient conclusions.

The net judgement: AI assistance is a genuine productivity multiplier for scaffolding, typesetting, and code boilerplate, but requires active, technically informed oversight wherever empirical accuracy matters.

ACKNOWLEDGEMENTS

Data are from the CDC/ATSDR 2024 Environmental Justice Index (<https://www.atsdr.cdc.gov/placeandhealth/eji/>). Analysis conducted in Python 3.11 with `pandas`, `statsmodels`, `xgboost`, `shap`, and `seaborn`. Completed for QSS 45 (AI and Machine Learning for Social Science), Dartmouth College, Spring 2026.

DATA AVAILABILITY

The EJI 2024 dataset is publicly available at <https://www.atsdr.cdc.gov/placeandhealth/eji/data.html>. Analysis notebooks are available upon request to the corresponding author.

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